P1. a) $g(E) = \sum_{\varepsilon_i} \delta(E - \varepsilon_i)$

Near K, K' points $\varepsilon_\pm(k) = \hbar v_F |\vec{k}| = v_F |\vec{p}|$

$$g(E) = \underset{\substack{\uparrow \\ \text{valley} \\ K, K'}}{2} \cdot \underset{\substack{\uparrow \\ \text{spin}}}{2} \sum_k \delta(E - \varepsilon_+(k)) + \delta(E - \varepsilon_-(k))$$

Assume $E < 0$, then $\delta(E - \varepsilon_+(k)) = \delta(-|E| - \hbar v_F |\vec{k}|) = 0$ because it can never be satisfied

$$\Rightarrow g(E)\Big|_{E<0} = 4 \sum_k \delta(E - \varepsilon_-(k)) = 4 \sum_k \delta(-|E| + \hbar v_F |\vec{k}|)$$

$$= \underset{\substack{\uparrow \\ \text{area}}}{4A} \int \frac{d\vec{p}}{(2\pi\hbar)^2} \delta(|E| - v_F p) = 4A \frac{2\pi}{(2\pi\hbar)^2} \cdot$$

$$\cdot \int_0^\infty dp \cdot p \cdot \delta(|E| - v_F p) = \frac{4A}{2\pi\hbar^2 v_F^2} \cdot |E|$$

let now $E > 0 \Rightarrow \delta(E - \varepsilon_-(k)) = \delta(E + \hbar v_F |k|) = 0$

$$\Rightarrow g(E)\Big|_{E>0} = 4 \sum_k \delta(E - \varepsilon_+(k)) =$$

$$= 4A \int \frac{d\vec{p}}{(2\pi\hbar)^2} \delta(E - v_F p) = \frac{4AE}{2\pi\hbar^2 v_F^2}.$$

Thus, for any sign of E

$$g(E) = \frac{2A}{\pi\hbar^2 v_F^2} |E| = \frac{g_s g_v A}{2\pi\hbar^2 v_F^2} |E|$$

(See p. 95 too)

b) See p. 94 of the lecture Notes: $C_V \sim A k_B \left(\frac{k_B T}{\hbar v_F}\right)^2$.

$$U(T\neq 0) = \underset{\substack{\uparrow \\ \text{spin degeneracy } g_s}}{2}A \int \frac{d\vec{k}}{(2\pi)^2}\left(E_+(k)\, f(E_+) + E_-(k)\, f(E_-)\right)$$

$$= \left(\begin{matrix}\text{see}\\ \text{p. 93}\end{matrix}\right) = 2Ag_s \int \frac{d\vec{k}}{(2\pi)^2} E_+(k)\, f(E_+) - g_s A \int \frac{d\vec{k}}{(2\pi)^2} E_+(k).$$

Approximate $E_+(\vec{k})$ by contributions from K, K' points

$$\Rightarrow U(T\neq 0) = 2\int_0^\infty dE\; g(E)\, E\, f(E) + U(T=0)$$

↳ integration over $E_+(\vec{k})$ domain

$$= \frac{2 g_s g_v A}{2\pi\hbar^2 v_F^2}\int_0^\infty dE\, \underset{\substack{\uparrow \\ \text{from } g(E)}}{E}\, E\, \frac{1}{e^{\beta E}+1} = \left(\frac{E}{k_B T} = x\right) =$$

$$= \frac{2 g_s g_v A}{2\pi\hbar^2 v_F^2}(k_B T)^3 \int_0^\infty dx\, \frac{x^2}{e^x+1}$$

$$= \frac{4A}{\pi} k_B T \left(\frac{k_B T}{\hbar v_F}\right)^2 \underbrace{\int_0^\infty \frac{dx\, x^2}{e^x+1}}_{\frac{3}{2}\zeta(3) \approx \frac{3}{2}\cdot(1.202)}$$

← this is exactly the result of p. 94

$$C_v = \frac{\partial U}{\partial T} \sim A k_B \left(\frac{k_B T}{\hbar v_F}\right)^2.$$

P2 Start with p. 82 of the notes:

original graphene Hamiltonian
$$H_0 = \sum_{k\in BZ} \left(c^\dagger_{A\vec{k}\sigma}, c^\dagger_{B\vec{k}\sigma}\right) \begin{pmatrix} 0 & \xi(k) \\ \xi^*(k) & 0 \end{pmatrix} \begin{pmatrix} c_{A\vec{k}\sigma} \\ c_{B\vec{k}\sigma} \end{pmatrix}$$

Then
$$\delta H = \sum_{k\in BZ} \left(c^\dagger_{Ak\sigma}, c^\dagger_{Bk\sigma}\right) \begin{pmatrix} m & 0 \\ 0 & -m \end{pmatrix} \begin{pmatrix} c_{Ak\sigma} \\ c_{Bk\sigma} \end{pmatrix}$$

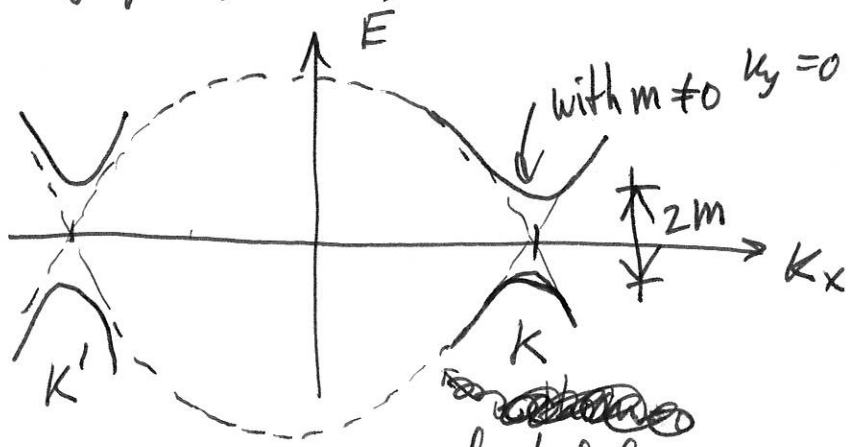
$$H = H_0 + \delta H = \sum_{k\in BZ} c^\dagger_{k\sigma} \begin{pmatrix} m & \xi(k) \\ \xi^*(k) & -m \end{pmatrix} c_{k\sigma}$$

spinor

Band structure follows from
$$\det \begin{pmatrix} m-E & \xi \\ \xi^* & -m-E \end{pmatrix} = 0$$

$$\Rightarrow \quad E^2 - m^2 - |\xi|^2 = 0, \quad E_\pm(\vec{k}) = \pm\sqrt{m^2 + |\xi|^2}$$

Since $\xi(\vec{k}) = 0$ for $\vec{k} = \vec{K} = \left(\frac{4\pi}{3a}, 0\right)$ and $\vec{k} = \vec{K}' = -\vec{K}$, we see that $m \neq 0$ leads to gap opening at these momenta



dashed line: $m = 0$.

$$c_{A/B\,\vec{k}\,\sigma} = \frac{1}{\sqrt{N_A}} \sum_{\vec{k}\in BZ} e^{i\vec{k}\cdot\vec{r}}\, c_{A/B\,\vec{k}\,\sigma}$$

But low-energy excitations live near K and K'

$\Rightarrow$ write Hamiltonian near these points + the rest:

$$H = \sum_{\vec{k}\in BZ} c^{+}_{k\sigma}\, \mathcal{H}(\vec{k})\, c_{k\sigma}$$

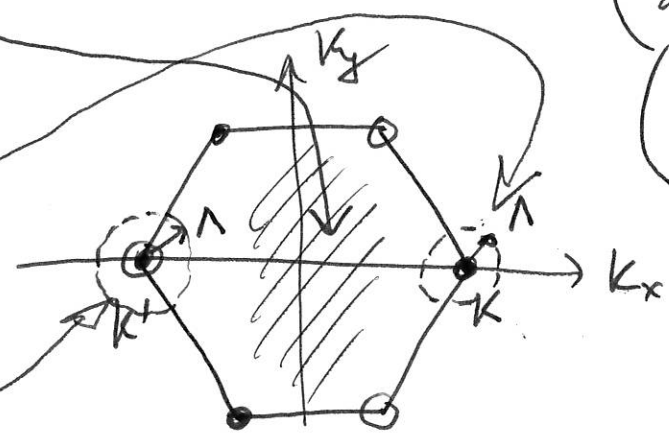
$$\mathcal{H}(\vec{k}) = \begin{pmatrix} m & \xi_k \\ \xi_k^{*} & -m \end{pmatrix}$$

$$H = \sum_{\vec{k}=\vec{K}+q}{}' \; c^{+}_{\vec{k}+\vec{q}\,\sigma}\, \mathcal{H}(\vec{K}+\vec{q})\, c_{\vec{k}+\vec{q}\,\sigma} \quad \left(\text{contribution from } \vec{k}\approx\vec{K}\right)$$

$$+ \sum_{\vec{k}=-\vec{K}+\vec{q}}{}' \; c^{+}_{-\vec{k}+\vec{q}}\, \mathcal{H}(-\vec{K}+\vec{q})\, c_{-\vec{k}+\vec{q}} \quad \left(\text{from } \vec{k}\approx\vec{K}'\right)$$

$$+ \sum_{\vec{k}}{}'' \; c^{+}_{k\sigma}\, \mathcal{H}(\vec{k})\, c_{k\sigma} \quad \leftarrow \left(\text{here } \vec{k} \text{ is far away from } K, K'\right)$$

(describes only high-energy processes)



$$H_{\text{low energy}} = \int_{q \le \Lambda} \frac{d\bar{q}}{(2\pi)^2} \, \psi^{+}_{K\sigma}(\bar{q}) \, \mathcal{H}_K(\bar{q}) \, \psi_{K\sigma}(\bar{q})$$

$$+ \int_{q \le \Lambda} \frac{d\bar{q}}{(2\pi)^2} \, \psi^{+}_{K'\sigma}(\bar{q}) \, \mathcal{H}_{K'}(\bar{q}) \, \psi_{K'\sigma}(\bar{q})$$

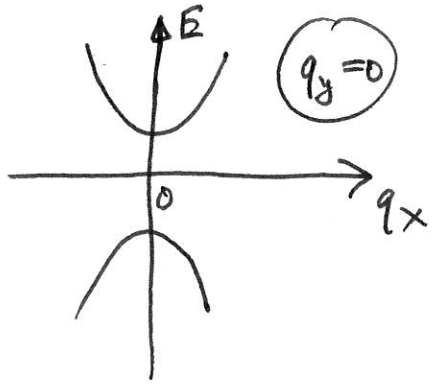
$$\begin{pmatrix} \psi_{AK\sigma}(\bar{q}) \\ \psi_{BK\sigma}(\bar{q}) \end{pmatrix} = \psi_{K\sigma}(\vec{q}) = c_{\vec{K}+\vec{q}\,\sigma} \quad \text{and similarly, } \psi_{K'\sigma}(\bar{q}) = c_{K'+q\,\sigma}$$

$$\mathcal{H}_K(\bar{q}) = \mathcal{H}(\vec{K}+\vec{q}) = \begin{pmatrix} m & \hbar v_F(q_x - i q_y) \\ \hbar v_F(q_x + i q_y) & -m \end{pmatrix}$$

$$\mathcal{H}_{K'}(\bar{q}) = \mathcal{H}(\vec{K}'+\bar{q}) = \begin{pmatrix} m & -\hbar v_F(q_x + i q_y) \\ -\hbar v_F(q_x - i q_y) & -m \end{pmatrix}$$

Thus near K: $E_\pm(\bar{q}) = E_\pm(\vec{K}+\vec{q}) = \pm\sqrt{m^2 + \hbar^2 v_F^2 (q_x^2 + q_y^2)}$

(and the same for K')



Note also that

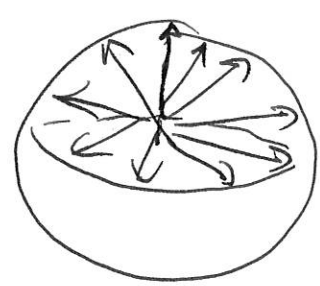
$$\mathcal{H}_K(\vec{q}) = m\sigma^z + \hbar v_F q_x \sigma^x + \hbar v_F q_y \sigma^y$$

$$\mathcal{H}_{K'}(\vec{q}) = +m\sigma^z \; \hbar v_F(-q_x)\sigma^x + \hbar v_F q_y \sigma^y$$

$\Rightarrow$ can define vectors $\begin{cases} \vec{h}_K(\vec{q}) = \hbar v_F(q_x\hat{x} + q_y\hat{y}) + m\hat{z} \\ \vec{h}_{K'}(\vec{q}) = \hbar v_F(-q_x\hat{x} + q_y\hat{y}) + m\hat{z} \end{cases}$

and unit vectors $\hat{n}_{K/K'}(\vec{q}) = \dfrac{\vec{h}_{K/K'}(\vec{q})}{\sqrt{\hbar^2 v_F^2(q_x^2+q_y^2)+m^2}}$

Since $m>0$, unit vectors $\hat{n}_{K/K'}(\vec{q})$ never enter southern hemisphere

$\Downarrow$

this gapped state is topologically trivial band insulator.